



NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY

An Autonomous Institution Approved by UGC/AICTE/Govt. of Karnataka

Accredited by NBA (Tier-I) and NAAC 'A+' Grade

Affiliated to Visvesvaraya Technological University, Belagavi

Post Box No. 6429, Yelahanka, Bengaluru-560064, Karnataka, INDIA



KNOWLEDGE • CHARACTER • UNITY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

21ADA581 WEB APPLICATION DEVELOPMENT

NAME	:	_____
REGISTER NO.	:	_____
YEAR/SEM	:	_____
SECTION	:	_____
BRANCH	:	_____

ASSIGNMENT NOTE BOOK

REGNO. _____

Certified that this is a bonafide observation of Practical work done by
Mr/Ms/Mrs.....of the.....
Semester..... Branch during the Academic
year.....in the.....laboratory.

Marks Allotted ____ Marks

Staff-in-Charge

Head of the Department

Internal Examiner

External Examiner

LIST OF ASSIGNMENT EXPERIMENTS

S.No	NAME OF THE EXPERIMENT
1.	Create an XHTML Page using tags to accomplish the following: - A Paragraph containing text “Life is Beautiful.” Bold face and italicize this text - Create Equation: $X = 1/3(y1^2 + z1^2)$ - Put a background image (any) to a page and demonstrate all attributes of background image - Create unordered list of 5 countries and ordered list of 3 countries
2.	Create following table using XHTML tags. Properly align cells, give suitable cell padding and cell spacing, and apply background colour, bold and emphasis necessary
3.	Use HTML5 to perform following tasks: - Draw a square using HTML5 SVG, fill the square with green colour and make 6px brown stroke width - Write the following mathematical expression using HTML5 MathML. $D = x^2 - y$ - Redirecting current page to another page after 5 seconds using HTML5 meta tag.
4.	Demonstrate the following HTML5 Semantic tags - for a webpage that gives information about educational institute
5.	Create a class called income, and make it background colour of #0ff. - Create a class called expenses, and make it background colour of #f0f. - Create a class called profit, and make it background colour of #f00. Throughout that document, and text that mentions income, expenses, or profit, attach the appropriate class to that piece of text, further create following line of text in the same document:
6.	Create following web page using HTML and CSS with tabular layout
7.	Create following calculator interface with HTML and CSS
8.	Write a java script program that on clicking a button, displays scrolling text which moves from left to right with a small delay
9.	Create a webpage containing 3 overlapping images using HTML, CSS and JS. Further when the mouse is over any image, it should be on the top and fully displayed

1. Create an XHTML Page using tags to accomplish the following:
 - A Paragraph containing text “Life is Beautiful.” Bold face and italicize this text
 - Create Equation: $X = 1/3(y1^2 + z1^2)$
 - Put a background image (any) to a page and demonstrate all attributes of background image
 - Create unordered list of 5 countries and ordered list of 3 countries

Code:

```
<!DOCTYPE html

PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">

<head>
  <title> Assessment Q1</title>
</head>

<body>
  <li> Q1 (1) A paragraph containing text “Life is Beautiful”. Bold face and italicize this text
  <p>
Life, a tapestry woven with threads of laughter and tears, whispers of wind and roars of thunder. It
dances in the sunrise hues, sings in the melodies of rain, and pirouettes under the twinkling canopy of
stars. Each beat of the heart, a brushstroke on the canvas, painting moments of joy, splashes of
sorrow, and shades of quiet contemplation. <br />
The beauty of life lies not just in grand gestures, but in the gentle brushstrokes of everyday moments.
The warmth of a sun-kissed cheek, the comfort of a shared smile, the silent understanding in a loved
one's gaze – these are the jewels that adorn our journey. Like sunlight dappling through leaves,
happiness flickers in fleeting moments – a child's giggle, the scent of freshly baked bread, the quiet
solace of a book. <br />
Yes, life has its storms, its days painted in shades of grey. But even in the midst of hardships, the
resilience of the human spirit shines through. It is a constant dance of learning, growing, and
evolving, a journey filled with breathtaking vistas and hidden paths waiting to be discovered. In the
quiet whisper of existence, <b><i>Life is Beautiful</i></b> - a treasure woven with joy and
sorrow.<br />
</p>
</li>

<li> Q1 (2) Creating the Equation <br />
<p>
 $x = 1/3(y1^{2} + z1^{2})$ 
</p>
</li>

<li> Q1 (3) Put a background image(any) to a page and demonstrate all attributes of background
image
<style type="text/css">
  body {
    background-image: url("https://imgs.search.brave.com/JX437uf1kSbCFe5AL2-
XDPq4HdjnhbdeFkiLQLxDV4Y/rs:fit:860:0:0/g:ce/aHR0cHM6Ly9pbWFn/ZXMudW5zcGxhc2gu/
Y29tL3Bob3RvLTB0/OTcyOTA3NTY3NjAtMjNhYzU1ZWVmZmZm/P3E9ODAmDz0xMDAw/J
mF1dG89Zm9ybWF0/JmZpdD1jcm9wJm14bGliPXJiLTQuMC4z/Jm14aWQ9TTN3eE1q/QTNmREI
4TUh4bGVl/QnNiM0psTFdabFpX/UjhNVFo4Zkh4bGJu/d3dmSHg4Zkh3PQ"); /* Set your image
URL here */
    background-repeat: no-repeat; /* Repeat horizontally */
    background-position: center; /* Position at top left corner */
  }
</style>
```

```

        background-attachment: scroll; /* Scroll with the page */
        background-size: cover; /* Scale to 50% of original size */
    }
</style>

```

<p>This page has a background image with various attributes applied.

```

<ul style="list-style-type: square;">
<li> Background-image: Sets the image URL. </li>
<li> Background-repeat: Controls repetition (e.g., repeat-x, repeat-y, no-repeat). </li>
<li> Background-position: Positions the image (e.g., top, bottom, center, left, right). </li>
<li> Background-attachment: Fixes the image to the viewport (fixed) or scrolls with the page (scroll).
</li>
<li> Background-size: Resizes the image (e.g., cover, contain, percentages) </li>
</ul>
</p></li>

```

```

<li> Q1 (4) Create unordered list of 5 countries and ordered list of 3 countries
<h2>Unordered List of Countries</h2>

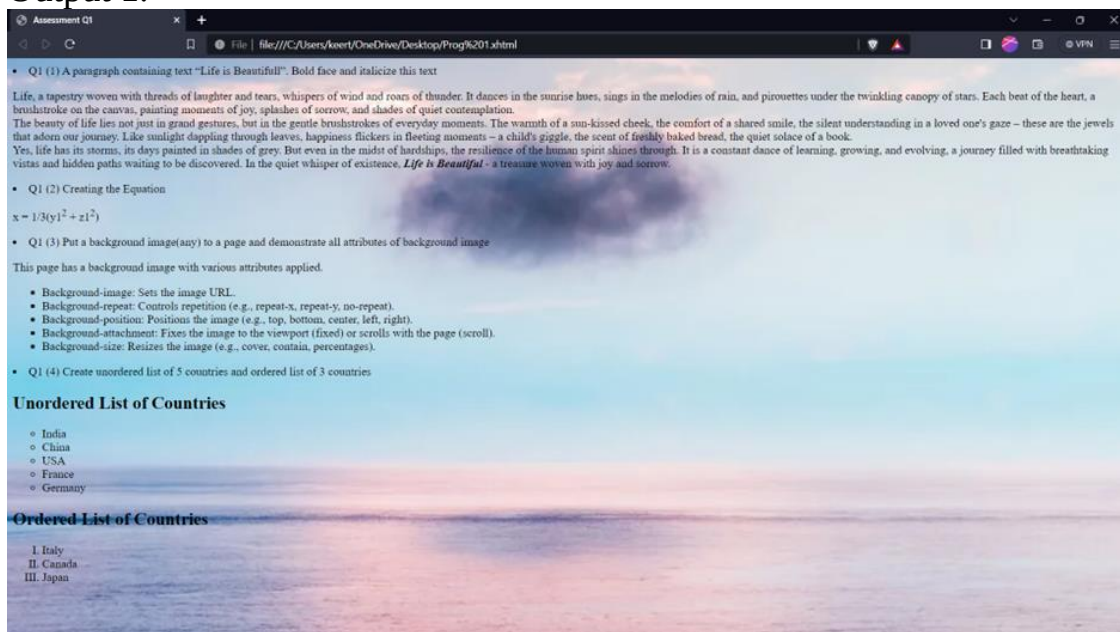
```

```

    <ul style="list-style-type: circle;">
        <li>India</li>
        <li>China</li>
        <li>USA</li>
        <li>France</li>
        <li>Germany</li>
    </ul>
<h2>Ordered List of Countries</h2>
    <ol style="list-style-type: Upper-Roman;">
        <li>Italy</li>
        <li>Canada</li>
        <li>Japan</li>
    </ol>
</li>
</body>
</html>

```

Output 1:



2. Create following table using XHTML tags. Properly align cells, give suitable cell padding and cell spacing, and apply background colour, bold and emphasis necessary

Code:

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en">
<head>
<title> XHTML Tables </title>
<meta name="author" content="Putta" />
<meta name="date" content="2023-02-17T04:58:37+0530" />
<meta http-equiv="content-type" content="text/html; charset=UTF-8"/>
<meta http-equiv="content-type" content="application/xhtml+xml; charset=UTF-8"/>
<meta http-equiv="content-style-type" content="text/css"/>
<meta http-equiv="expires" content="0"/>
<link rel="stylesheet" type="text/css" href="style.css" />
</head>
<body>

<style>
table, th, td {
    border: 1px solid black;
    border-collapse: collapse;
}
th, td{
    padding-left: 10px;
    padding-bottom: 20px
}
table {
    border-spacing: 30px;
}
</style>

<h3>Table using XHTML tags</h3>

<table align="center" style="width: 70%; height: 450px;">
<tr>
<td rowspan="9" align="center" bgcolor="#DEEEEE">
<b>Department</b>
</td>
<td rowspan="3" align="center" bgcolor="#9E7BA0">
<b>Sem1</b>
</td>
<td style="padding: 15px;">
<em>SubjectA</em>
</td>
</tr>
<tr>
<td ><em>SubjectB</em></td>
</tr>
<tr>
<td ><em>SubjectC</em></td>
</tr>
<tr>
<td rowspan="3" align="center" bgcolor="#9E7BA0">
<b>Sem2</b>
```

```

</td>
  <td ><em>SubjectE</em></td>
</tr>
<tr>
  <td ><em>SubjectF</em></td>
</tr>
<tr>
  <td ><em>SubjectG</em></td>
</tr>

<tr>
  <td rowspan="3" align="center" bgcolor="#9E7BA0">
    <b>Sem3</b>
  </td>
  <td ><em>SubjectH</em></td>
</tr>
<tr>
  <td ><em>SubjectI</em></td>
</tr>
<tr>
  <td ><em>SubjectJ</em></td>
</tr>
</table>

</body>
</html>

```

Output 2:



Table using XHTML tags

Department	Sem1	SubjectA
		SubjectB
		SubjectC
	Sem2	SubjectE
		SubjectF
		SubjectG
	Sem3	SubjectH
		SubjectI
		SubjectJ

3. Use HTML5 to perform following tasks:
- Draw a square using HTML5 SVG, fill the square with green colour and make 6px brown stroke width
 - Write the following mathematical expression using HTML5 MathML. $D = x^2 - y^2$
 - Redirecting current page to another page after 5 seconds using HTML5 meta tag.

Code:

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8" />
    <title>HTML5 Demo</title>

</head>
<body>
    <li> Q3 (1)
        <h3> HTML5 SVG: </h3>
        <svg width="200" height="200" align="centre">
            <rect x="50" y="50" width="100" height="100" fill="green" stroke="brown" stroke-
width="6px"/>
        </svg>
    </li>

    <li> Q3 (2)
        <h3> HTML5 MathML Equation: </h3>

        <math xmlns = "http://www.w3.org/1998/Math/MathML">
            <mrow>
                <msup><mi>d</mi></msup>
                <mo> = </mo>
                <msup><mi>x</mi><mn>2</mn></msup>
                <mo>-</mo>
                <msup><mi>y</mi><mn>2</mn></msup>
            </mrow>
        </math>
    </li>
    <br>

    <li> Q3 (3)
        <h3> Redirecting Page: </h3>

        <meta http-equiv="refresh" content="5; url=https://www.nmit.ac.in/">
        <title>Redirecting...</title>

        This web page is developed for demonstrating the redirecting process!.... <br>
        You will be redirected to NITTE Home page in 5 seconds.
    </li>

</body>
</html>
```

Output 3:

HTML5 Demo

File | file:///C:/Users/keert/OneDrive/Desktop/Prog%203.html

- Q3 (1)

HTML5 SVG:



- Q3 (2)

HTML5 MathML Equation:

$$d = x^2 - y^2$$

- Q3 (3)

Redirecting Page:

This web page is developed for demonstrating the redirecting process!....
You will be redirected to NITTE Home page in 5 seconds.

Before redirecting the page

Output 3: Redirected Page goes to NMIT home page

Best Engineering Colleges in Bangalore

https://www.nmit.ac.in

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An Autonomous Institution With An Grade By MAAC UGC | Approved By UGC / AICTE / Govt. of Karnataka | Bangalore

CREDENTIAL VERIFICATION ADMISSIONS ACHIEVEMENTS PLACEMENT CONTACT US

ABOUT US PROGRAMS RESEARCH LIFE@NMIT QUICK LINKS DISCOVER NITTE

ADMISSIONS OPEN 2024-25

Engineering Applications Open
2024-25
@NMIT, Bangalore
(Admission via. Online Entrance Test)
APPLY NOW

SPECIAL SCHOLARSHIP
for Meritorious Students*

APPLY NOW

PARENT PORTAL
E-RESULTS
Tech - QnA

After redirecting the page

4. Demonstrate the following HTML5 Semantic tags - for a webpage that gives information about educational institute

Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Edu Institution</title>

<style>

    body{

        background-color: #FFFDD0;

    }

</style>

</head>

<body>

<header>

<h1 style="text-align: center; color: #3498DB;">Brigade Global University</h1>

<nav>

<ul>

    <li><a href="#">Home</a></li>

    <li><a href="#">Programs</a></li>

    <li><a href="#">Admissions</a></li>

    <li><a href="#">About Us</a></li>

    <li><a href="#">Contact</a></li>

</ul>

</nav>

</header>

<main>

<article>

    <h2>Welcome to Global University</h2>

    <p>Beyond brick and mortar, a university hums with the symphony of ambitions. Within its walls, curiosity ignites minds like sparks in tinder, setting alight journeys of discovery across myriad disciplines.

    Here, classrooms transcend mere lecture halls, transforming into vibrant forums where intellectual debates dance on the airwaves. Libraries, brimming with the whispers of the past, become treasure
```

troves to unearth knowledge long buried. Laboratories, like alchemists' workshops, bubble with experimentation, pushing the boundaries of known and unknown. It's a refuge for the restless soul, a playground for the inquisitive mind, and a launchpad for dreams that dare to reach for the stars.

</p>

</article>

<section>

<h2>Featured Programs</h2>

<figure>

<dl>

<dt>CSE<dd>Computer Science & Engineering

<dt>ISE<dd>Information Science & Engineering

<dt>AI & DS<dd>Artificial Intelligence and Data Science

<dt>AI & ML<dd>Artificial Intelligence and Machine Learning

<dt>ECE<dd>Electronics & Communication Engineering

</dl>

</figure>

<p>Ignite your Passion and Discover your calling and unleash your hidden talents within our diverse academic landscape. Join a vibrant community: Be part of a dynamic network of scholars, professors, and peers who share your thirst for knowledge.</p>

<p><mark>For further enquires Please do Contact Us Ph:080-2345678 </mark></p>

</section>

<aside>

<h3>Quick Links</h3>

Apply Now

Schedule a Tour

Request Information

Quick Chat


```

</aside>

</main>

<footer>

    <p style="text-align: center;">&copy; 2024 Brigade Global University</p>

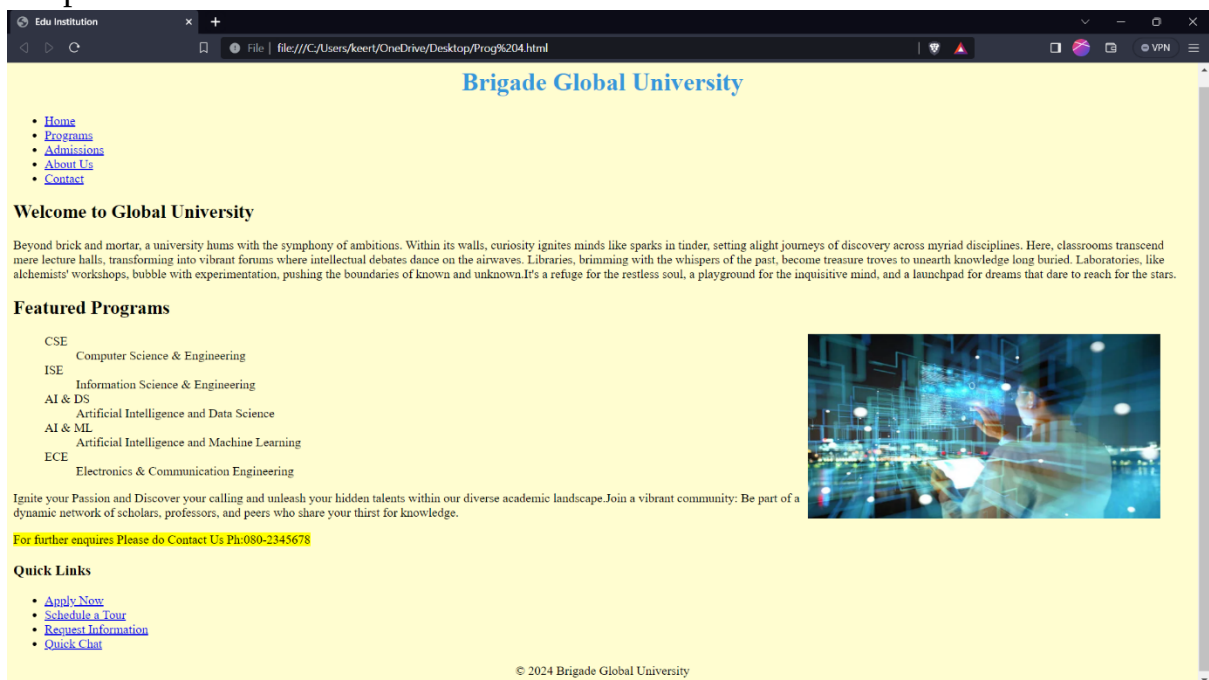
</footer>

</body>

</html>

```

Output 4:



- Create a class called income, and make it background colour of #0ff.**

 - Create a class called expenses, and make it background colour of #f0f.**
 - Create a class called profit, and make it background colour of #f00.**

Throughout that document, and text that mentions income, expenses, or profit, attach the appropriate class to that piece of text, further create following line of text in the same document:

Code:

```

<!DOCTYPE html>

<html>

<head>

<meta charset="utf-8" />

```

```
<title>Class Selectors</title>
```

```
<style>
```

```
.income {background-color: #0ff; font-style: italic;}
```

```
.expenses {background-color: #f0f;font-style: oblique;}
```

```
.profit {background-color: #ff0;font-weight: bold;}
```

```
.red{color: red;font-size: 24px;}
```

```
.strike{text-decoration: line-through; font-size: 24px;}
```

```
p {font-family: Cursive;}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Selecting the appropriate class</h1>
```

```
<p>
```

From groceries to growth, managing income, expenses, and profit keeps your personal or business finances on track.

```
</p>
```

```
<p class="income">
```

Income: Individuals earn income through salaries, investments, gigs, or selling goods and services, while businesses bring in money through sales, investments, or rentals.

Predictable paychecks or unpredictable bonuses, income can be a steady stream or a burst of financial sunshine..

```
</p>
```

```
<p class="expenses">
```

Expenses: Rent, salaries, materials – fixed costs underpin daily life and business operations. Then come variable costs like groceries and production supplies. Finally, discretionary choices like travel and entertainment round out the spending picture.

Expenses can be predictable staples like rent and salaries, or fluctuate with activity like materials and travel, while others, like entertainment, add bursts of color to the financial canvas.</p>

```
<p class="profit">
```

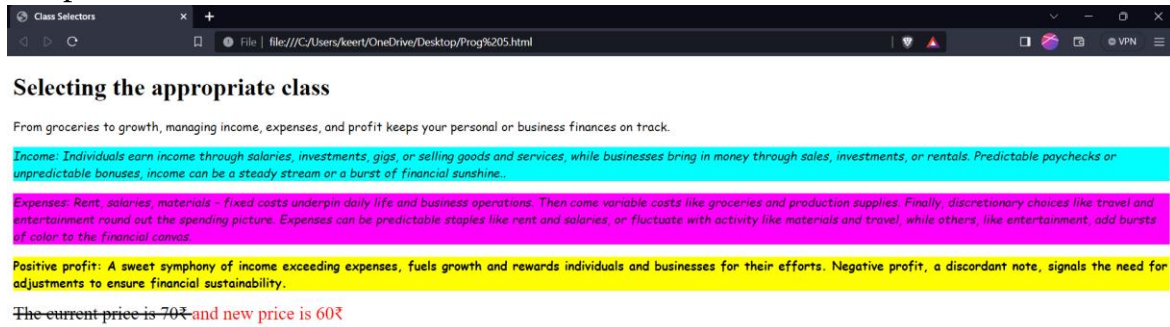
Positive profit: A sweet symphony of income exceeding expenses, fuels growth and rewards individuals and businesses for their efforts. Negative profit, a discordant note, signals the need for adjustments to ensure financial sustainability.</p>

```
<span class="strike">The current price is 70₹ </span><span class="red">and new price is 60₹</span>
```

```
</body>
```

</html>

Output 5:



6. Create following web page using HTML and CSS with tabular layout

Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>Sign Up</title>

    <style>

        body {

            font-family: Arial, sans-serif;

            background-color: #f2f2f2;

        }

        .container {

            width: 500px;

            margin: 0 auto;

            padding: 20px;

            background-color: #F7E7CE;
```

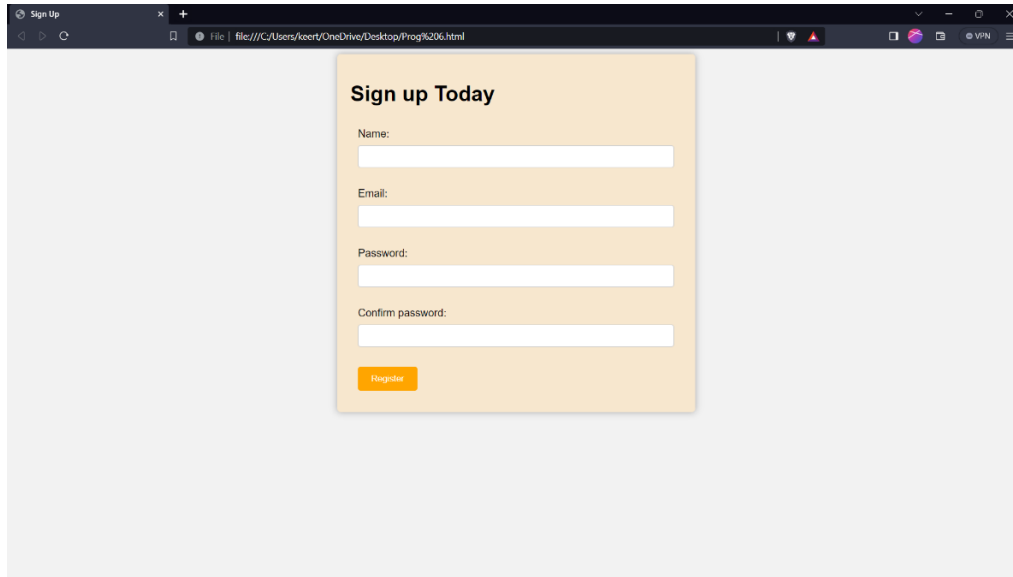
```

        border-radius: 5px;
        box-shadow: 0 0 10px rgba(0,0,0,0.3);
    }
    table {
        width: 100%;
    }
    th, td {
        padding: 10px;
    }
    th {
        text-align: left;
        background-color: #f2f2f2;
    }
    input[type=text], input[type=password], input[type=email] {
        width: 100%;
        padding: 8px;
        margin: 8px 0;
        border: 1px solid #ccc;
        border-radius: 4px;
        box-sizing: border-box;
    }
    button[type=submit] {
        background-color: #FFA500;
        color: #fff;
        padding: 10px 20px;
        border: none;
        border-radius: 4px;
        cursor: pointer;
    }
</style>
</head>
<body>

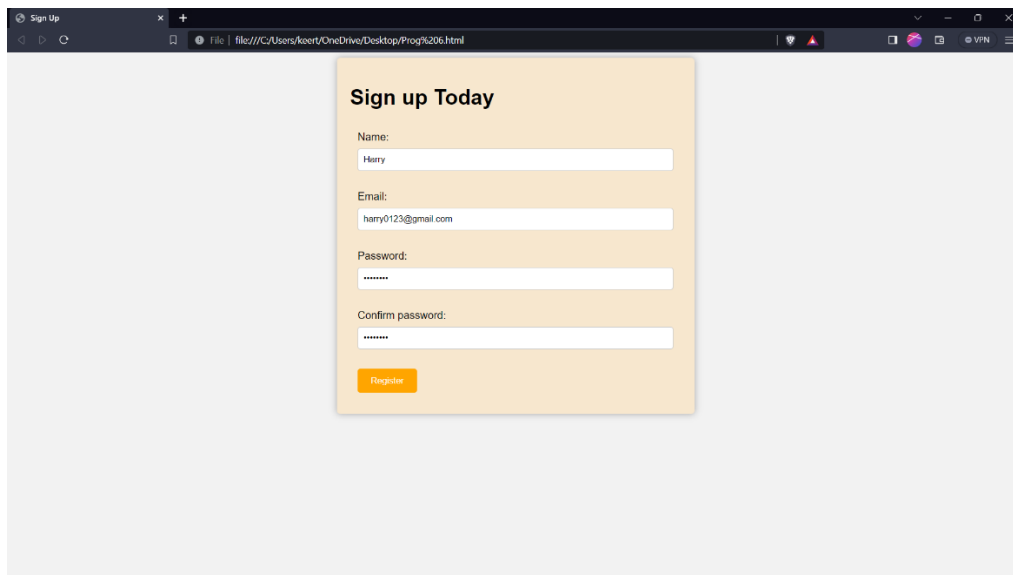
```

```
<div class="container">
  <h1>Sign up Today</h1>
  <form>
    <table>
      <tr>
        <td><label for="username">Name:</label> <br>
          <input type="text" id="username" name="username"
required></td>
      </tr>
      <tr>
        <td><label for="email">Email:</label> <br>
          <input type="email" id="email" name="email"
required></td>
      </tr>
      <tr>
        <td><label for="password">Password:</label> <br>
          <input type="password" id="password" name="password"
required></td>
      </tr>
      <tr>
        <td><label for="password">Confirm password:</label>
          <br>
          <input type="password" id="password" name="password"
required></td>
      </tr>
      <tr>
        <td colspan="2"><button
type="submit">Register</button></td>
      </tr>
    </table>
  </form>
</div>
</body>
</html>
```

Output 6:



A screenshot of a web browser window displaying a 'Sign up Today' form. The form is centered on a light gray background. It features four input fields: 'Name:', 'Email:', 'Password:', and 'Confirm password:'. Below these fields is an orange 'Register' button. The browser's address bar shows the file path: file:///C:/Users/keert/OneDrive/Desktop/Prog%206.html.



A second screenshot of the same 'Sign up Today' form, now populated with sample data. The 'Name' field contains 'Harry', the 'Email' field contains 'harry0123@gmail.com', the 'Password' field contains '*****', and the 'Confirm password' field contains '*****'. The orange 'Register' button is still at the bottom. The browser's address bar remains the same.

7. Create following calculator interface with HTML and CSS

Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Calculator</title>
```

```
<style>

body {
  font-family: Arial, sans-serif;
  background-color: #e6e6fa;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
  margin: 0;
}

#calculator {
  background-color: #F0C0FF;
  border-radius: 10px;
  padding: 20px;
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.2);
}

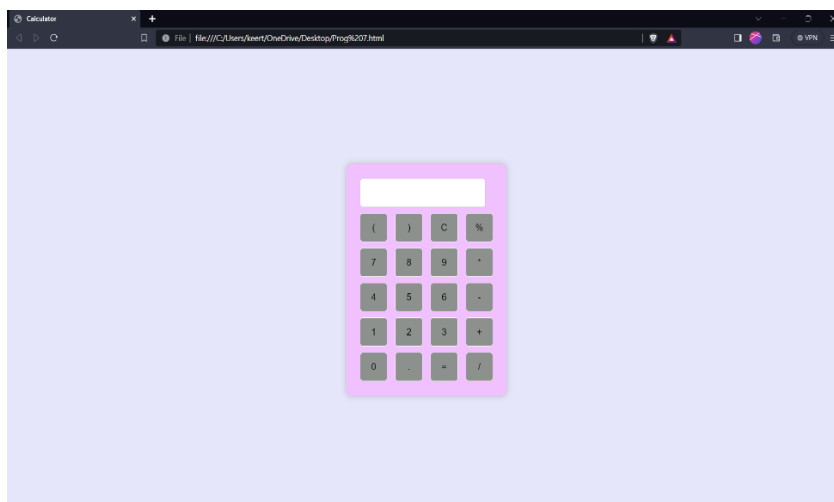
button {
  width: 50px;
  height: 50px;
  margin: 5px;
  font-size: 16px;
  background-color: #8D918D;
  border: 1px solid #fff;
  border-radius: 5px;
  cursor: pointer;
}

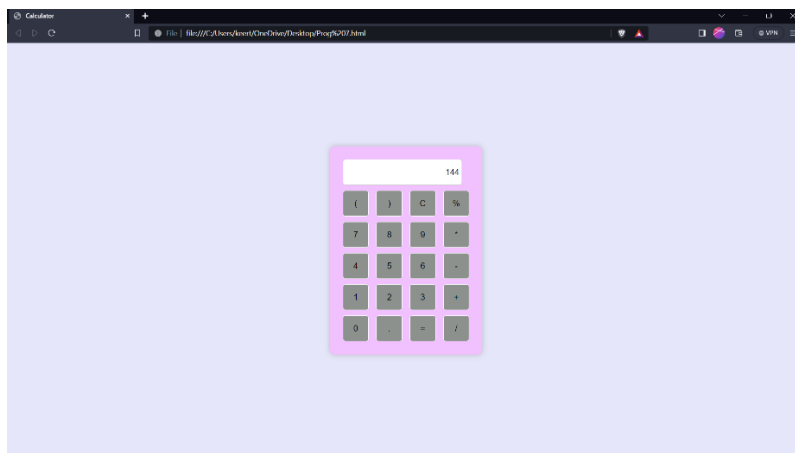
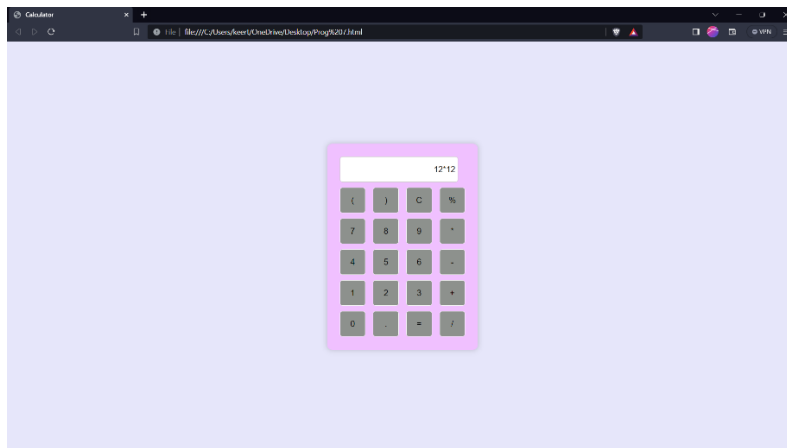
input {
  width: 230px;
  height: 50px;
  margin: 5px;
  font-size: 16px;
  border: 1px solid #ccc;
```

```
border-radius: 5px;
padding: 5px;
box-sizing: border-box;
text-align: right;
}
</style>
</head>
<body>
<div id="calculator">
<input type="text" id="display" readonly>
<br>
<button onclick="appendCharacter('(')">(</button>
<button onclick="appendCharacter(')')">)</button>
<button onclick="clearDisplay()">C</button>
<button onclick="appendCharacter('%)">%</button>
<br>
<button onclick="appendCharacter('7')">7</button>
<button onclick="appendCharacter('8')">8</button>
<button onclick="appendCharacter('9')">9</button>
<button onclick="appendCharacter('*)">*</button>
<br>
<button onclick="appendCharacter('4')">4</button>
<button onclick="appendCharacter('5')">5</button>
<button onclick="appendCharacter('6')">6</button>
<button onclick="appendCharacter('-)">-</button>
<br>
<button onclick="appendCharacter('1')">1</button>
<button onclick="appendCharacter('2')">2</button>
<button onclick="appendCharacter('3')">3</button>
<button onclick="appendCharacter('+)">+</button>
<br>
<button onclick="appendCharacter('0')">0</button>
```

```
<button onclick="appendCharacter('.')">.</button>
<button onclick="calculateResult()">=</button>
<button onclick="appendCharacter('/')">/</button>
</div>
<script>
function appendCharacter(character) {
document.getElementById('display').value += character;
}
function clearDisplay() {
document.getElementById('display').value = "";
}
function calculateResult() {
try {
document.getElementById('display').value =
eval(document.getElementById('display').value);
} catch (error) {
document.getElementById('display').value = 'Error';
}
}
</script>
</body>
</html>
```

Output 7:





8. Write a java script program that on clicking a button, displays scrolling text which moves from left to right with a small delay

Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Scrolling Text Example</title>

<style>

#scrollingText {

font-size: 24px; font-weight: bold;

white-space: nowrap; overflow: hidden;

}

</style>

</head>

<body>

<button id="startBtn">Start Scrolling</button>
```

```
<div id="scrollingText">This is some scrolling text!</div>

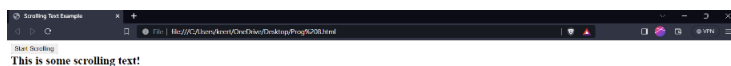
<script>

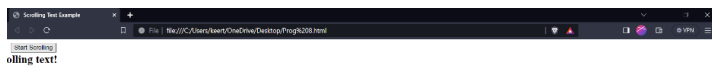
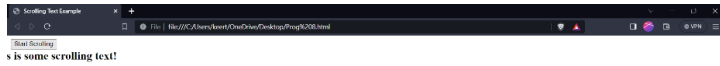
  var scrollingText = document.getElementById("scrollingText");
  var startBtn = document.getElementById("startBtn");
  var textWidth = scrollingText.clientWidth;
  var containerWidth = scrollingText.parentNode.clientWidth;
  var currentPosition = 0;
  function scrollText() {
    if (currentPosition < containerWidth) {
      scrollingText.style.transform = "translateX(-" + currentPosition + "px)";
      currentPosition += 1;
      setTimeout(scrollText, 20);
    } else {
      currentPosition = -textWidth;
      scrollText();
    }
  }
  startBtn.addEventListener("click", function() {
    currentPosition = 0;
    scrollText();
  });
</script>

</body>

</html>
```

Output 8:





-
- 9. Create a webpage containing 3 overlapping images using HTML, CSS and JS. Further when the mouse is over any image, it should be on the top and fully displayed**

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Overlapping Images</title>

<style>

body {

margin: 0;

padding: 0;

overflow: hidden;

}

.image-container {

position: absolute;
```

```
transition: z-index 0.3s ease-in-out;
}
img {
width: 300px;
height: auto;
position: absolute;
cursor: pointer;
transition: transform 0.3s ease-in-out;
}
img:hover {
transform: scale(1.2);
z-index: 1;
}
#image1 { top: 100px; left: 100px; }
#image2 { top: 200px; left: 200px; }
#image3 { top: 300px; left: 300px; }
</style>
</head>
<body>
<div class="image-container" id="image1">
  
</div>
<div class="image-container" id="image2">
  
</div>
<div class="image-container" id="image3">
  
</div>
</body>
</html>
```

Output 9:

